

Polynomial Regression Assignment

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This report describes the polynomial regression models used for the two personalized datasets. The objective was to predict the continuous target y while respecting the degree limits in the assignment.

1. Data and preprocessing

The var1 dataset contains six input variables (x_1 - x_6), while var2 contains three (x_1 - x_3). Each training file contains 1,000 rows and a target column y ; each test file contains 1,000 rows and the matching input columns. The supplied data are already numeric and cleaned, with values bounded approximately in $[-1, 1]$. No target information or rows from one variant were used for the other. `PolynomialFeatures` expands each row into all interaction and power terms whose total degree is at most d . The expanded columns are standardized before ordinary least-squares linear regression to improve numerical conditioning.

2. Degree selection

For each allowed degree, I used five-fold cross-validation with shuffling and random seed 42. The same folds were used for R^2 and MSE. R^2 was the primary selection criterion because it measures the fraction of target variation explained; MSE was retained as a scale-dependent diagnostic. This procedure evaluates generalization rather than choosing the degree from training error, which would favor unnecessarily flexible polynomials.

Variant	Degree range	Chosen degree	Mean CV R^2	Mean CV MSE
var1	1-10	4	0.9331	0.6799
var2	1-20	8	0.9928	0.2878

The selected degree for var1 is 4. Its validation R^2 improves through degree 4 and then drops, indicating that higher-degree terms begin to overfit the available samples. For var2, the validation score improves through degree 8 and declines at degree 9 and above. Degree 8 therefore provides the best bias-variance trade-off among the permitted choices.

3. Final fitting and inference

After selecting the degree, I refit the complete pipeline on all 1,000 training rows for that variant and predicted the corresponding 1,000 test rows. The final files contain one column, y , and preserve test-row order, as required by `sample_submission.csv`. The script is deterministic: paths are resolved relative to `solve.py`, the cross-validation seed is fixed, and no random augmentation or external data is used.

4. Deliverables and reproducibility

Training and inference code: `solve.py`. Dependencies: `requirements.txt`. Generated prediction files: `BT2024259_pred_var1.csv` and `BT2024259_pred_var2.csv`. The repository is available at https://github.com/Kandi-Rohan/ML_assignment. Running `python solve.py` from the repository root regenerates both files.